

Supplemental Material for

“Topological sound in non-reciprocal active-liquid networks”

Minqi Cheng^{1#}, Jingming Chen^{1#}, Linyun Yang^{1†}, Zhen Gao^{1*}

¹State Key Laboratory of Optical Fiber and Cable Manufacturing Technology, Department of Electronic and Electrical Engineering, Guangdong Key Laboratory of Integrated Optoelectronics Intellisense, Southern University of Science and Technology, Shenzhen 518055, China

[#]These authors contributed equally: Minqi Cheng and Jingming Chen.

^{†}Corresponding author. Email: yanglinyun@cqu.edu.cn (L.Y.Y.); gaoz@sustech.edu.cn (Z.G.)*

Supplemental Note 1: Particle-based simulation for the steady-state chiral flows

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Movies S1-S2: Simulated steady-state chiral flows

Note 1: Particle-based simulation for the steady-state chiral flows

To confirm the validity of the steady-state chiral flow considered in Fig. 1(b) in the main text, we use the following particle model [1-2] to perform the particle-based simulation,

$$m\dot{\mathbf{v}}_i = -\gamma\mathbf{v}_i + F_0 \left(\hat{\mathbf{v}}_i + \sum_{\langle \mathbf{x}_i, \mathbf{x}_j \rangle} \frac{\hat{\mathbf{v}}_i}{N} \right) + \sum_j \nabla_i U(|\mathbf{x}_i - \mathbf{x}_j|) + \sqrt{2\gamma k_B T} \hat{\zeta}_i(t), \quad (\text{S1})$$

where $\mathbf{x}_i$ and $\mathbf{v}_i$ are the position and velocity of each particle and $\hat{\mathbf{v}}_i$ is the unit vector parallel to $\mathbf{v}_i$, $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ denotes the neighbors of the i th particle and N is the number of neighboring particles, $U(|\mathbf{x}_i - \mathbf{x}_j|)$ represents repulsive potential between particles to avoid the condensation, and $\sqrt{2\gamma k_B T} \hat{\zeta}_i(t)$ is the white Gaussian noise term. Here, we use Yukawa potential $U(r) = \frac{b}{re^{\kappa r}}$ and a steep one-sided harmonic repulsive potential $k_p x^2/2$ to implement the boundaries of annular channels and coupling tubes. By performing numerical simulation of this model starting from the designed initial state, we observe the emergence of steady-state chiral flow in annular channels whose direction depends on the initial state. Inside each coupling tube between adjacent annular channels, the active particles choose different small channels according to the direction of movement. Fig. S1 shows two typical snapshots of the global steady-state chiral flows formed by self-propelled particles. These results justify the existence of the global steady-state chiral flow in Fig. 1(b). The present particle-based model can be described by the Toner-Tu equations in the hydrodynamic limit.

We refer to this annular channel filled with active liquid as a circulator, and each circulator is connected by three tubes to construct a topological active-liquid honeycomb network, as illustrated in Figs. 1(a)-(b).

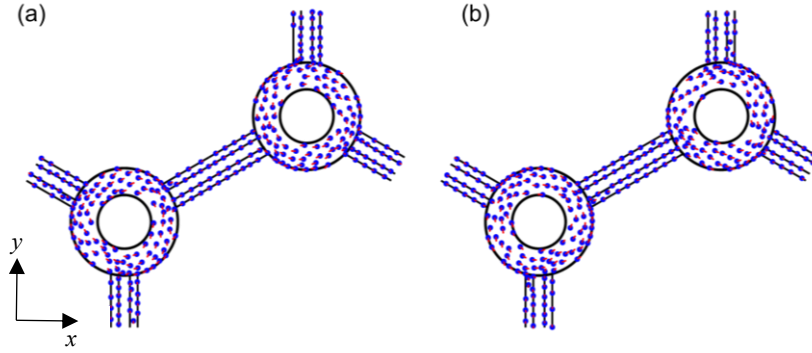


Fig. S1. The steady-state chiral flows formed by self-propelled particles in a honeycomb network of annular channels. (a)-(b) A unit cell of the topological active-liquid network with periodic boundary conditions along the x and y directions. Each blue dot represents an active particle, and a red bar extending from it indicates the velocity of each particle. The total number of particles is 300 and the parameters are $m = \gamma = 1$, $F_0 = 0.4$, $b = 5$, $\kappa = 0.1$, $k_p = 50$, and $k_B T = 2 \times 10^{-4}$. The direction of the flow can be decided by the initial conditions.

Note 2: Derivation of the Schrödinger-like equation

In this section, we show the details of the derivation of the equations. The full Toner-Tu equations read

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (\text{S2})$$

$$\begin{aligned} & \partial_t \mathbf{v} + \lambda(\mathbf{v} \cdot \nabla) \mathbf{v} + \lambda_2(\nabla \cdot \mathbf{v}) \mathbf{v} + \lambda_3 \nabla |\mathbf{v}|^2 = \\ & (\alpha - \beta |\mathbf{v}|^2) \mathbf{v} - \nabla P + D_B \nabla(\nabla \cdot \mathbf{v}) + \gamma \nabla^2 \mathbf{v} + D_2(\mathbf{v} \cdot \nabla)^2 \mathbf{v} + \mathbf{f}, \end{aligned} \quad (\text{S3})$$

where $\rho(\mathbf{r}, t)$ is the density field of active matter and $\mathbf{v}(\mathbf{r}, t)$ is the local average of velocities of self-propelled particles. Eq. (S2) represents the equation of continuity. In Eq. (S3), the first term of its right-hand side suggests a preference for a nonzero constant speed $|\mathbf{v}_0| = \sqrt{\alpha/\beta}$ if α is positive. D_B, γ, D_2 are the diffusion constants (or viscosities). The $\mathbf{f}$ term is a random driving force representing the noise. The coefficients in these equations can be obtained from microscopic models. Neglecting the D_B, D_2 terms and λ_2, λ_3 terms for the sake of simplification, we can obtain their simplest form read:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (\text{S4})$$

$$\partial_t \mathbf{v} + \lambda(\mathbf{v} \cdot \nabla) \mathbf{v} = (\alpha - \beta |\mathbf{v}|^2) \mathbf{v} - \frac{c^2}{\rho_0} \nabla \rho + \gamma \nabla^2 \mathbf{v}, \quad (\text{S5})$$

where we assume that the equation of state $P = c^2 \rho / \rho_0$ with c being the sound speed and that the external force $\mathbf{f}$ is absent. These equations are exactly the Eqs. (1)-(2) in the main text. We consider the fluctuations of density and velocity fields from a steady state, $\delta \rho(\mathbf{r}, t) = \rho(\mathbf{r}, t) - \rho_0$, $\delta \mathbf{v}(\mathbf{r}, t) = \mathbf{v}(\mathbf{r}, t) - \mathbf{v}_0$, and neglect the dissipative term $\gamma \nabla^2 \mathbf{v}$. We will discuss the effects of this dissipative term in the case of non-Hermitian perturbations in Supplementary Note 8. Substituting $\delta \rho$ and $\delta \mathbf{v}$ into the Eqs. (S4) - (S5), we obtain:

$$\partial_t(\delta \rho + \rho_0) + \nabla \cdot [(\delta \rho + \rho_0)(\delta \mathbf{v} + \mathbf{v}_0)] = 0 \quad (\text{S6})$$

and

$$\partial_t(\delta \mathbf{v} + \mathbf{v}_0) + \lambda[(\delta \mathbf{v} + \mathbf{v}_0) \cdot \nabla](\delta \mathbf{v} + \mathbf{v}_0) = (\alpha - \beta |\delta \mathbf{v} + \mathbf{v}_0|^2)(\delta \mathbf{v} + \mathbf{v}_0) - \frac{c^2}{\rho_0} \nabla(\delta \rho + \rho_0). \quad (\text{S7})$$

Neglecting the second-order contributions of $\delta \rho$ and $\delta \mathbf{v}$, we then obtain:

$$\begin{aligned} & \partial_t \delta \rho + \nabla \cdot (\delta \rho \mathbf{v}_0 + \rho_0 \delta \mathbf{v} + \rho_0 \mathbf{v}_0) \\ & = \partial_t \delta \rho + \nabla \delta \rho \cdot \mathbf{v}_0 + \rho_0 \nabla \cdot \delta \mathbf{v} = 0 \end{aligned} \quad (\text{S8})$$

and

$$\begin{aligned} \partial_t \delta \mathbf{v} + \lambda(\mathbf{v}_0 \cdot \nabla) \delta \mathbf{v} & = |\mathbf{v}_0|^2 \beta (\delta \mathbf{v} + \mathbf{v}_0) - \beta (|\mathbf{v}_0|^2 + 2\delta \mathbf{v} \cdot \mathbf{v}_0)(\delta \mathbf{v} + \mathbf{v}_0) - \frac{c^2}{\rho_0} \nabla \delta \rho \\ & = -2\beta (\delta \mathbf{v} \cdot \mathbf{v}_0) \mathbf{v}_0 - \frac{c^2}{\rho_0} \nabla \delta \rho. \end{aligned} \quad (\text{S9})$$

The final equations read:

$$\partial_t \delta \rho + (\mathbf{v}_0 \cdot \nabla) \delta \rho = -\rho_0 \nabla \cdot \delta \mathbf{v} \quad (\text{S10})$$

$$\partial_t \delta \mathbf{v} + \lambda(\mathbf{v}_0 \cdot \nabla) \delta \mathbf{v} = -2\beta (\mathbf{v}_0 \cdot \delta \mathbf{v}) \mathbf{v}_0 - \frac{c^2}{\rho_0} \nabla \delta \rho. \quad (\text{S11})$$

We can also derive the Schrödinger-like equation from Eqs. (S10)-(S11). Applying the convective derivative $\partial_t + \lambda(\mathbf{v}_0 \cdot \nabla)$ to Eq. (S10),

$$\begin{aligned}
[\partial_t + \lambda(\mathbf{v}_0 \cdot \nabla)][\partial_t \delta\rho + (\mathbf{v}_0 \cdot \nabla)\delta\rho] &= -[\partial_t + \lambda(\mathbf{v}_0 \cdot \nabla)]\rho_0 \nabla \cdot \delta\mathbf{v} \\
&= -[\rho_0 \partial_t (\nabla \cdot \delta\mathbf{v}) + \lambda(\mathbf{v}_0 \cdot \nabla)(\rho_0 \nabla \cdot \delta\mathbf{v})] = -[\rho_0 \nabla \cdot (\partial_t \delta\mathbf{v}) + \rho_0 \lambda \nabla \cdot ((\mathbf{v}_0 \cdot \nabla)\delta\mathbf{v})] \\
&= -\rho_0 \nabla \cdot [\partial_t \delta\mathbf{v} + \lambda(\mathbf{v}_0 \cdot \nabla)\delta\mathbf{v}] = \rho_0 \nabla \cdot \left(2\beta(\mathbf{v}_0 \cdot \delta\mathbf{v})\mathbf{v}_0 + \frac{c^2}{\rho_0} \nabla \delta\rho \right) = \rho_0 \nabla \cdot \frac{c^2}{\rho_0} \nabla \delta\rho \\
&= c^2 \nabla^2 \delta\rho.
\end{aligned}$$

That is

$$[\partial_t + \lambda(\mathbf{v}_0 \cdot \nabla)][\partial_t + (\mathbf{v}_0 \cdot \nabla)]\delta\rho = c^2 \nabla^2 \delta\rho \quad (\text{S12})$$

and then we assume an oscillating solution $\delta\rho(\mathbf{r}, t) = \delta\tilde{\rho}(\mathbf{r})e^{i\omega t}$,

$$\begin{aligned}
[\partial_t + \lambda(\mathbf{v}_0 \cdot \nabla)][\partial_t + (\mathbf{v}_0 \cdot \nabla)]\delta\tilde{\rho} &= (i\omega + \lambda v_{0,i} \partial_i)(i\omega + v_{0,i} \partial_i)\delta\tilde{\rho} \\
&= [-\omega^2 + i\omega(1 + \lambda)v_{0,i} \partial_i + \lambda v_{0,i}^2 \partial_i^2]\delta\tilde{\rho} = c^2 \partial_i^2 \delta\tilde{\rho}.
\end{aligned}$$

Under the condition that $v_0 \ll c$, we have

$$[\omega^2 - i\omega(1 + \lambda)v_{0,i} \partial_i + c^2 \partial_i^2]\delta\tilde{\rho} = 0, \quad (\text{S13})$$

or

$$\left[(\nabla - i\mathbf{A})^2 + \frac{\omega^2}{c^2} \right] \delta\tilde{\rho} = 0 \quad (\text{S14})$$

where $\mathbf{A} \equiv \omega(\lambda + 1)\mathbf{v}_0/(2c^2)$. This show that the velocity field acts as an effective vector potential for the propagation of density waves.

Note 3: Topological invariant of the non-reciprocal active-liquid network

One can define a Chern number for each Floquet band n by integrating the Berry curvature associated with the Bloch eigenstate $\{|\psi_n(\mathbf{k})\rangle\}$ over the first Brillouin zone (BZ) [5]. Fig. S2 shows the distributions of Berry curvatures of the six Floquet bands ($n = 1, \dots, 6$) of the AFI [Fig. S2(a)] and CI [Fig. S2(b)] phases, as well as their Chern number, respectively. It can be seen that all Chern numbers of the Floquet bands of the AFI are zero. Therefore, the Chern number cannot fully characterize the nontrivial properties of the non-reciprocal active-liquid network since the AFI possesses topological edge states despite the Chern numbers of all surrounding bands being zero. Here we adopt a “winding number” to characterize the topological properties of the non-reciprocal active-liquid network, which is an integer quantized topological invariant defined by [5]

$$W_\phi(U_\phi) = \frac{1}{8\pi^2} \int_{[0,T] \times \text{BZ}} dt dk_x dk_y \text{Tr} \left(U_\phi^{-1} \partial_t U_\phi \left[U_\phi^{-1} \partial_{k_x} U_\phi, U_\phi^{-1} \partial_{k_y} U_\phi \right] \right), \quad (\text{S15})$$

where ϕ is a chosen value of the quasi-energy in a spectral gap of the evolution operator U , or $S(\mathbf{k})$ in our case, and therefore W_ϕ is termed a gap invariant which directly gives the number of chiral edge states in a given quasi-energy gap. The calculation of this invariant involves integrals over k_x , k_y , and the time variable t . In the context of network models, there is no meaningful definition of the evolution operator U for intermediate time t . In practice, one can define any $U(\mathbf{k}, t)$, such that $U(\mathbf{k}, T)$ is the evolution operator for the network. The choice is non-unique but will not affect the value of W_ϕ [6]. Here, we

adopt the interpolation method from Ref. [7],

$$\mathcal{U}_{int}(t) \equiv \exp(-itH_{-\pi}^{\text{eff}}) \quad (\text{S16})$$

with

$$H_{\theta}^{\text{eff}} = \frac{i}{T} \ln_{-\theta} U(T, \mathbf{k}). \quad (\text{S17})$$

We perform numerical integration to calculate W_{ϕ} , and their values converge to 1, 1, 1, 1, 1, 1 (1, 0, 1, 1, 0, 1) for the anomalous (Chern) bandgaps.

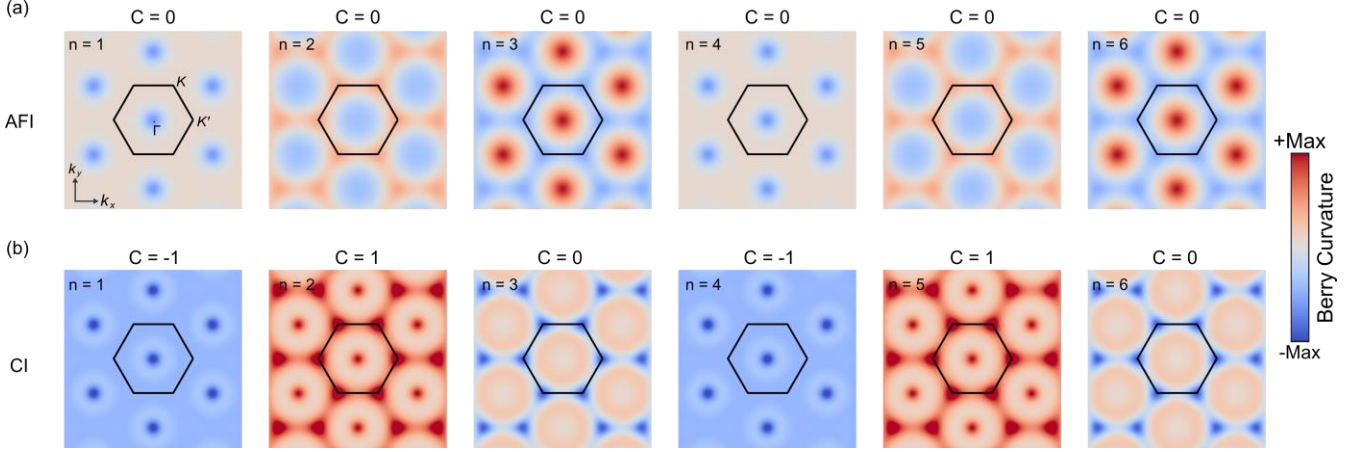


Fig. S2. The Berry curvature distributions for the (a) anomalous and (b) Chern phases. $n = 1, \dots, 6$ is the band index ordered by the quasi-energy ϕ from $-\pi$ to π .

Note 4: COMSOL Multiphysics simulations

We solve Eq. (S14) using the commercial finite element software COMSOL Multiphysics. The mesh is sufficiently fine to make sure that the results are convergent. In practice, we employ the coefficient PDE interface of the COMSOL, whose general form is

$$e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot (-c \nabla u - \alpha u + \gamma) + \beta \cdot \nabla u + au = f. \quad (\text{S18})$$

The coefficients of Eq. (S18) can be determined by expanding Eq. (S14) and comparing it with Eq. (S18) term by term. The boundary conditions used in our simulation are like sound hard boundary (i.e., the normal components of the gradients of the fields are zero, $\vec{n} \cdot \nabla u = 0$) everywhere except the periodic boundary conditions.

In our honeycomb active-liquid network, we have performed the particle-based numerical simulation (Supplementary Note 1) for a sufficient time for the dynamics to relax to a steady state. In the analysis performed for the density-wave computations, we take a simple form of the steady state as shown in Fig. S1. In the annular channel, we postulate the velocity field has uniform magnitude everywhere and is oriented azimuthally, that is, perpendicular to the vector connecting the position of the fluid to the nearest annulus center. In the region of each coupling tube, we set the net effective vector potential to zero since there are two opposite flows in it.

Note 5: Validation of the model assumptions

The density waves considered in our model is governed by the Schrödinger-like equation,

$$[(\nabla - i\mathbf{A})^2 + \omega^2/c^2]\delta\tilde{\rho} = 0, \quad (\text{S19})$$

where the vector potential $\mathbf{A} = \omega(\lambda + 1)\mathbf{v}_0/(2c^2)$ corresponds to the time-reversal symmetry breaking. By solving Eq. (S19) using the finite element method, we can obtain the scattering matrix S_M of a three-port circulator. To obtain the entries of S_M , we put a source at each port turn by turn and measure the complex amplitudes of the transmitted waves at all three ports. Meanwhile, we can use the coupled mode theory to obtain the scattering-wave model and the parameterized form of the scattering matrix [3,4],

$$S_0 = \begin{bmatrix} R & T & D \\ D & R & T \\ T & D & R \end{bmatrix}, \quad (\text{S20})$$

where $R = -1 + \frac{2}{3}\cos\xi e^{i\xi} + \frac{2}{3}\cos\eta e^{i\eta}$, $T = \frac{2}{3}\cos\xi e^{-i\frac{2}{3}\pi+i\xi} + \frac{2}{3}\cos\eta e^{i\frac{2}{3}\pi+i\eta}$, and $D = \frac{2}{3}\cos\xi e^{i\frac{2}{3}\pi+i\xi} + \frac{2}{3}\cos\eta e^{-i\frac{2}{3}\pi+i\eta}$, which includes only two angle variables $\xi, \eta \in (-\frac{\pi}{2}, \frac{\pi}{2}]$. Note that S_0 is a unitary matrix with C_3 symmetry. By varying the two angle variables, we can explore a family of circulators with all possible degrees of non-reciprocity.

To examine the validation of the model assumption, we first check the C_3 symmetry and the unitarity of the simulated scattering matrix S_M . The C_3 symmetry of the scattering matrix indicates that $S_{21} = S_{32} = S_{13}$ and $S_{11} = S_{22} = S_{33}$. As shown in Fig. S3(a), we plot the moduli and arguments of these quantities in the operating frequency range, which unambiguously show that the C_3 symmetry is a valid assumption. Next, we check the unitarity of S_M . Fig. S3(b) shows the eigenvalues of the simulated scattering matrix in the complex plane for various frequencies. We can see that they almost lie on the unit

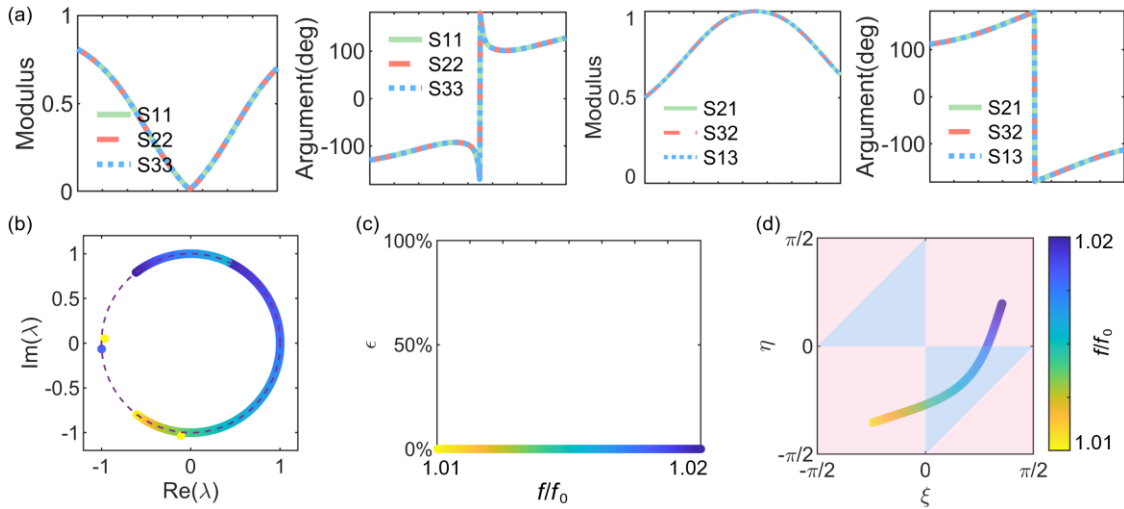


Fig. S3. Validation of the model assumption. (a) C_3 symmetry holds when $S_{11} = S_{22} = S_{33}$, and $S_{21} = S_{32} = S_{13}$ in the operating frequency range. (b) Eigenvalues of the scattering matrix S_M . (c) The error produced by approximating the simulated scattering matrix with Eq. (S20). (d) The track of positions (ξ, η) with the operating frequency varying from $1.01f_0$ to $1.02f_0$. $f_0 = c/2\pi r$ is the characteristic frequency, with r represents the average radius of the inner radius and the outer radius of the around channel and c is speed of sound.

circle, indicating that unitarity is also a very reasonable assumption. Finally, we estimate the error between the simulated and theoretical matrices by considering the Frobenius norm of their difference $S_M - S_0$. To do this, we extract (ξ, η) from S_M with almost C_3 symmetry and construct the theoretical scattering matrix S_0 according to Eq. (S20). We plot $\varepsilon = \|S_M - S_0\|_F$ in Fig. S3(c), in which we can see that the error is below 1% in the operating frequency range. Therefore, the model assumptions are valid within the scope of our discussion. Moreover, we plot the track of (ξ, η) in the complex plane as a function of the operating frequency in Fig. S3(d), indicating that the change of operating frequency will induce a topological phase transition.

Note 6: Disorder preparation

In this section, we consider the disorder stemming from the fluctuations of the particle speed and the geometrical parameters (inner radius) of the circulator. These two factors can be mapped to a region on the (ξ, η) plane, as indicated by the colored dots [the dot color (size) represents the particle speed (inner radius of the circulator)] in Figs. S4(a)-(b), respectively. This region represents the accessible (ξ, η) parameters in our finite element simulations. We select a fixed point on the (ξ, η) plane for both the anomalous [Fig. S4(a)] and the Chern [Fig. S4(b)] phases, and then consider various triangular regions centered on this point with increasing areas, which correspond to different disorder strengths with $\Delta_s = 0, 10\%$, and 25% , respectively. For each disorder strength, the parameters (particle speed and inner radius) of each circulator of a 6×6 finite sample are randomly selected from the corresponding triangular regions.

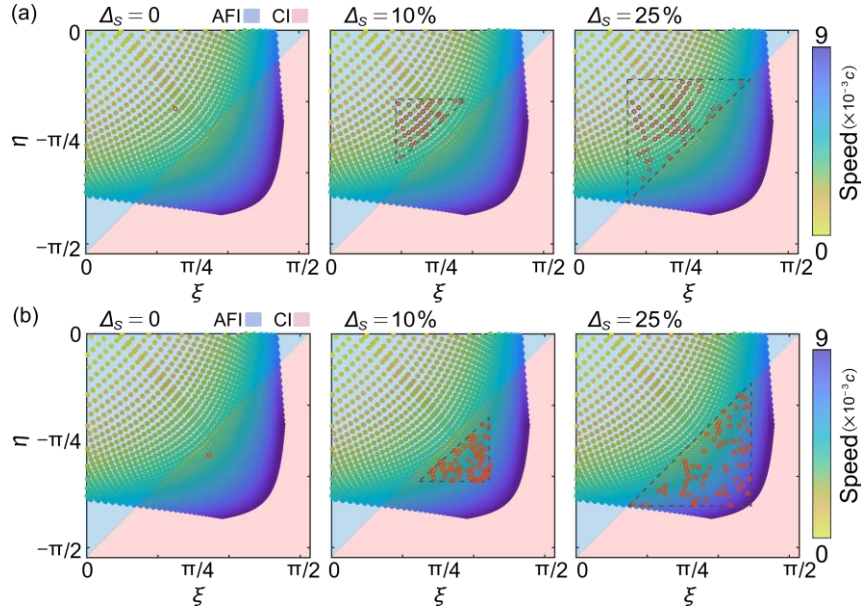


Fig. S4. The implementation of scattering matrix disorder. (a) Mapping of particle speed (dot color) and circulator inner radius (dot size) on the (ξ, η) plane. The blue (pink) region represents the anomalous (Chern) phase. The red circles are the values of random (ξ, η) used to introduce the disorder. (b) The same as (a), but for the Chern phase.

Note 7: The calculation of the non-Hermitian winding number

In the non-Hermitian active-liquid network, the non-Hermitian winding number (topological invariant)

can be calculated using the formula:

$$v_\alpha(\phi_0) = \int_{BZ} \frac{dk_\alpha}{2\pi} \frac{d}{dk_\alpha} \arg[e^{-i\phi(k_\alpha)} - e^{-i\phi_0}]. \quad (S21)$$

Note that since the periodicity of the real part of the phase factor ϕ , the eigen spectrum of ϕ lies on the surface of a cylinder with infinite length [i.e., $\text{Im}(\phi) \in \mathbb{R}$]. Hence, it possesses a nontrivial winding if $\text{Re}(\phi)$ varies from 0 to 2π throughout the BZ. However, this nontrivial winding originates from the geometric features of the cylinder (i.e., the periodicity of the Floquet spectra). To fully characterize both the winding in a 2D plane and around a cylinder, we map the surface of the cylinder to a 2D infinite plane by considering the value of $e^{-i\phi}$ instead of ϕ , as shown in Fig. S5(a). Consequently, a non-trivial winding of ϕ around a reference point ϕ_r is thus mapped to that of $e^{-i\phi}$ around $e^{-i\phi_r}$. On the other hand, the winding around the cylinder varies from $-\pi$ to π , corresponding to a non-trivial winding of $e^{-i\phi}$ around $e^{-i\phi_r} = 0$. Therefore, by properly choosing ϕ_r for the reference point, both of these two types of winding can be characterized by a winding number defined by Eq. (S21). Here, we adopt the method mentioned in Refs. [8,9], then the non-Hermitian winding number can be equivalently calculated using the formula:

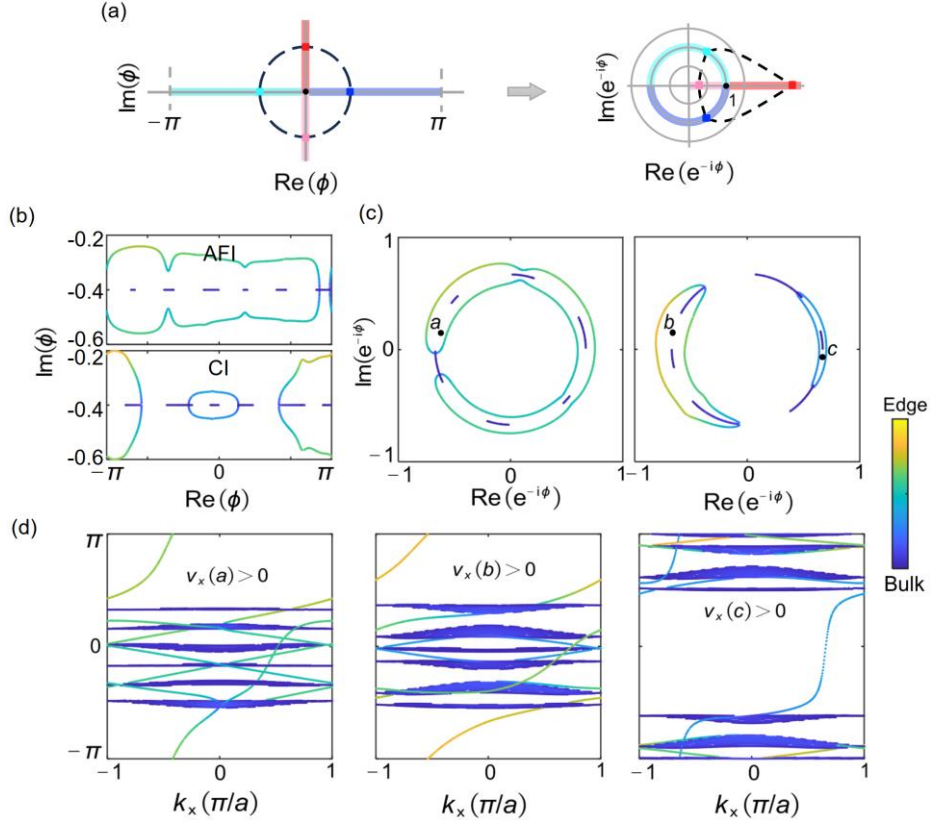


Fig. S5. The non-Hermitian winding number. (a) Mapping between the complex plane of ϕ (left panel) with $\text{Re}(\phi)$ having a period of 2π , and an infinite complex plane of $e^{-i\phi}$ (right panel). (b) The complex eigen spectrum of ϕ of AFI (upper panel) and CI (down panel) phases. All parameters are the same as those in Figs. 4(b)-(c). (c) The transformed complex eigen spectrum of AFI (left panel) and CI (right panel) phases on the complex plane of $e^{-i\phi}$. The black dots represent the reference points utilized to calculate the winding number. (d) Phase of the eigenvalues $e^{-i\phi}$ as a function of the wave vector k_x . When the reference point falls within the loop, the system exhibits a full spectral flow, indicating non-trivial winding. The states in (b)-(d) are color-coded based on the contribution of the edge states, revealing that the spectral flow is predominantly influenced by the edge states.

$$v_\alpha(\phi_0) = \sum_{n=1}^N \int_{BZ} \frac{dk_\alpha}{2\pi} \partial_{k_\alpha} \arg[e^{-i\phi_n(k_\alpha)} - e^{-i\phi_0}], \quad (\text{S22})$$

where N represents the total number of states, encompassing both bulk and edge states. We can alternatively plot the argument $\arg[e^{-i\phi_n(k_\alpha)} - e^{-i\phi_0}]$ and determine how many times the phase undergoes a complete spectral flow. In other words, how many times it smoothly traverses across the entire feasible range $[0, 2\pi)$ within the BZ. Here we calculate the winding number of the corner-localized chiral edge states [Figs. 4(b)-(c) or Fig. S5(c)]. If the reference points lie inside the loop of the edge states (e.g., black points a , b and c) and the complex eigenfrequencies exhibit a full spectral flow, then the phase has a nontrivial winding number.

Note 8: Robustness of the anomalous and Chern edge states against non-Hermitian perturbations

Now we consider the non-Hermitian case. Note that the dissipative term $\gamma \nabla^2 \mathbf{v}$ in Eq. (S3) suggests that the dissipation in the system is related to the velocity field of self-propelled particles. If the variation of the local average velocities $\mathbf{v}$ in a certain region changes rapidly, it will result in dissipation in this region. It has been reported that one can put passive particles in the microchannel to make the active matter (bacteria) flow and mutate in a certain region [10]. This may be an effective method to introduce tunable loss into the active matter systems.

Next, we explore the correlation between the robustness of anomalous and Chern edge states against non-Hermitian perturbations and their degrees of boundary localization at $g = 0$ (without loss). We find that at $g = 0$, the anomalous and Chern edge states exhibit different levels of boundary localization, as shown in Figs. S6(a)-(b). To quantitatively describe the levels of boundary localization of the anomalous and Chern edge states, we define and calculate a boundary localization factor $\sigma = \sum_i^n m_i |\psi_i|$ shown in Fig. S6(c), where m_i is the weight factor that decays from the edge to the bulk, and $|\psi_i|$ is the amplitude at the site indicated by the white dashed line in Fig. S6(a). Fig. S6(d) displays the fitting slopes of the curves shown in Fig. 4(e). Comparing Fig. S6(c) to S6(d), we find that the more tightly localized edge state (with a larger value of the strength of boundary localization) has a steeper decay coefficient slope against the loss factor g (with a smaller negative value of slope), indicating that the strength of boundary localization of the edge states directly influences their resistance to the non-Hermitian perturbations.

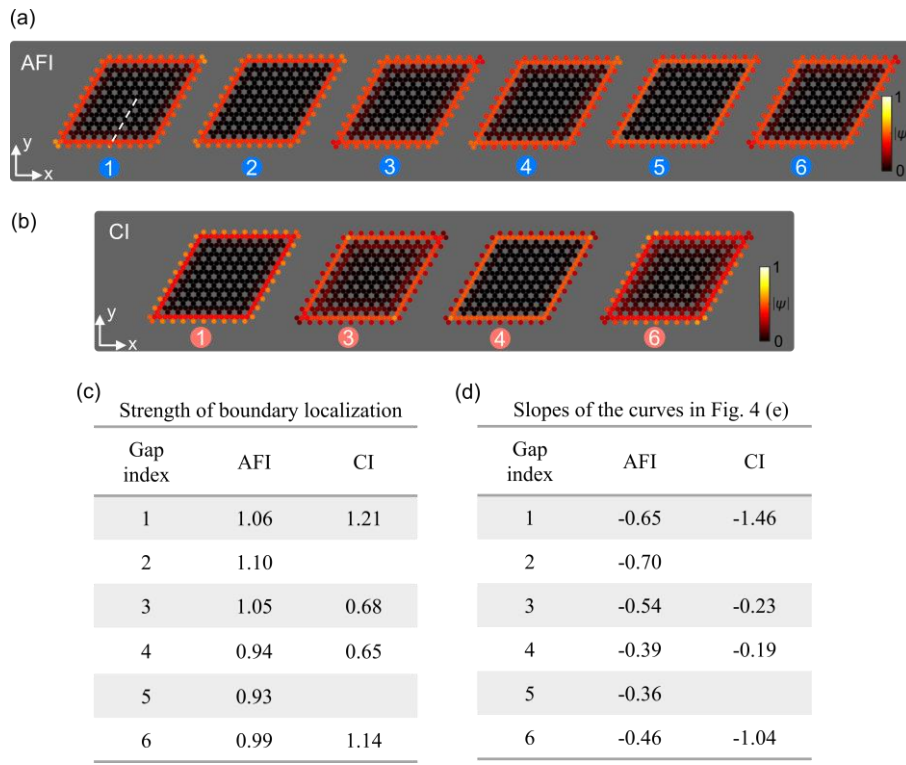


Fig. S6. Robustness comparison of AFI and CI edge states against non-Hermitian perturbations. (a-b) The mid-gap edge state profiles in different gaps for (a) AFI and (b) CI phases with gain (loss) factor $g = 0$. The number represents the gap index in Figs. 1(d)-(f). (c) The boundary localization strength of AFI and CI edge states in different gaps with $g = 0$. (d) The slopes of the decay coefficients as a function of the gain (loss) factor g shown in Fig. 4(e).

Movie S1-S2 (separate files): Particle-based numerical simulations of the dynamics of the self-propelled particles. Movie S1 shows the clockwise steady-state chiral flows formed by self-propelled particles, and Movie S2 shows the counter-clockwise steady-state chiral flows.

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